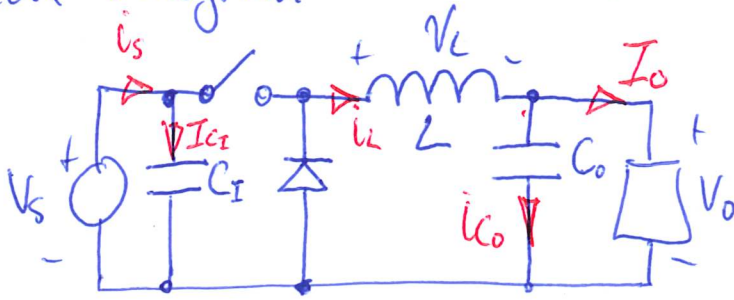
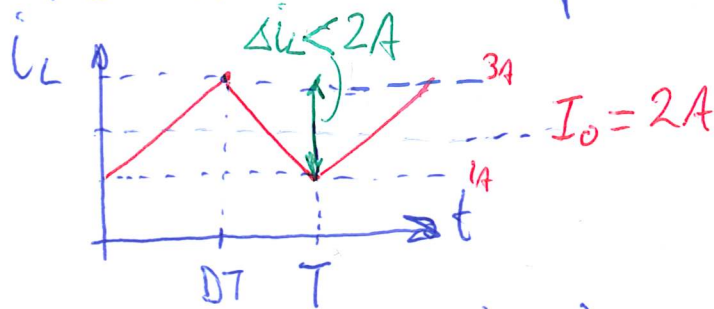


Buck Converter end-of-lecture Problem Solution.

Circuit diagram with labels



From the definition in the problem of Δi_L , we get



showing limit of maximum allowed current ripple for problem,

So $I_o > \Delta i_L$ (not $\Delta i_L/2$), giving

$$L > \frac{V_o (1-D)}{f_s I_o} = \frac{8(1-0.667)}{80 \times 10^3 \times 2}$$

$$D = \frac{V_o}{V_s} = \frac{8}{12} = 0.667$$

$$= 16.7 \mu H$$

So $L > 17 \mu H$

C_{out}? $\Delta V_o = V_o \left(\frac{1-D}{8LCf^2} \right)$ so $C_{out} > \frac{V_o}{\Delta V_o} \left(\frac{1-D}{8Lf^2} \right)$ where $\Delta V_o = 0.08 V$.

$$> \frac{8}{0.08} \left(\frac{1-0.667}{8 \times 16.7 \times 10^{-6} \times (80 \times 10^3)^2} \right)$$

$$> \underline{\underline{39 \mu F}}$$

C_{in}? $C_{in} > \frac{I_o D (1-D)}{f_s \Delta V_s}$ where $\Delta V_s = 0.12 V$.

$$> \frac{2 \times 0.667 (0.333)}{80 \times 10^3 \times 0.12}$$

$$> \underline{\underline{46 \mu F}}$$